

Project 2024

Digit identification through features extracted from audio signals

Background: Currently, human voice recognition and interpretation is available on practically all devices. For example, it is common for cell phones/computers to perform tasks activated by voice commands. The initial phase of any recognition task involves acquiring and extracting features from the audio signal. The main objective is that these characteristics allow different sounds to be discriminated, corresponding to different voice commands.

Objective: This project aims to analyze audio signals in the frequency and time domains with the aim of identifying the digits in English between 0 and 9.

Introduction¹: Figure 1 shows an illustration of the human speech production system. The main anatomical components of this system are the lungs, trachea, larynx (speech production organ), pharyngeal cavity (throat), oral cavity and nasal cavity. There are also other anatomical components that contribute to speech production, such as the vocal folds (or cords), tongue, lips, teeth, and jaw. The latter are referred to as articulators and move to different positions in order to produce various speech sounds such as vowels, consonants or other phonological units of a language.

The human speech production process can be summarized as follows. During speech, the air expelled from the lungs goes to the trachea and rises to the glottis, where the flow is periodically interrupted by the movement of the vocal cords. The tension of the vocal cords is adjusted by the larynx causing the cords to vibrate. Within the vocal tract, constrictions are also produced to the passage of air through the tongue, lips and nose. These changes in the configurations of the human speech system will cause the vibrations emitted through the lips into the surrounding air (i.e., the sound) to have different frequency components.

Speech production can be seen as a filtering operation, in which the three main cavities of the speech production system constitute the main acoustic filter.

¹ https://link.springer.com/referenceworkentry/10.1007/978-0-387-73003-5_199

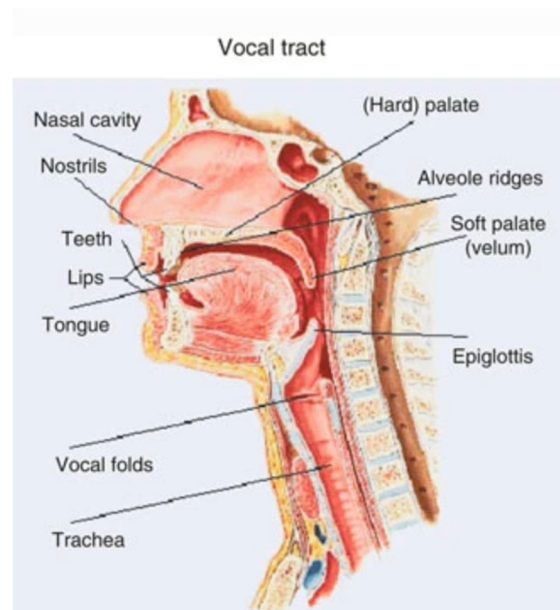


Fig 1: Main components of the human speech production system.

(Ref: https://link.springer.com/referenceworkentry/10.1007/978-0-387-73003-5_199)

Programming Language: **MATLAB (alternatively Python).**

Organization: **Groups of two students (whenever possible)**

Data: <https://www.kaggle.com/datasets/sripaadsrinivasan/audio-mnist/download?datasetVersionNumber=1>
(you must unzip the file and consider only the raw signals available in the *data* folder).

Data description: The provided data corresponds to voice signals emitted by 60 participants (each subdirectory in the *data* folder contains the signals corresponding to one participant). Each participant repeated each of the digits 50 times, meaning each of the 60 folders contains 500 audio signals in .wav format. Each audio signal was acquired at a sampling rate of 48000 Hz in single-channel mode. More details about the data can be obtained at:

<https://github.com/soerenab/AudioMNIST>

<https://www.kaggle.com/datasets/sripaadsrinivasan/audio-mnist>

<https://arxiv.org/abs/1807.03418>.

Evaluation Method:

The project was divided in four milestones, whose deadlines are:

Milestone 1: 03/03/2024, 23:59

Milestone 2: 08/04/2024, 23:59

Milestone 3: 05/05/2024, 23:59

Milestone 4: 19/05/2024, 23:59

On the deadline set for the completion of each milestone, the group must submit the corresponding work in the inforestudante.

If the group fails to meet the designated submission date, they may include the work in the submission of the following milestone, with penalties applied to the grades of the corresponding milestones as follows:

Delay of 1 *deadline*: 25 %

Delay of 2 *deadlines*: 50 %

Delay of 3 *deadlines*: 75 %

Materials to submit: It is highly recommended to submit **code**, **figures**, and **text** (questions asked throughout the exercises) in an integrated manner. For this, you can use tools such as Jupyter Notebook (for MATLAB or Python) or MATLAB's Live Script.

Defences:

- Final Defence: taking place during the week of **May 20th to 24th**, during PL hours or at times arranged with the PL's professor.
- Additional evaluations will be conducted in the PL classes after the submission of each milestone.
- The grade for the defence and participation in PL classes will multiply the grade obtained for the implemented work.

$$\text{Project Grade} = (30\% \text{ Report} + 70\% \text{ Code}) \times \text{Defence (0 - 100\%)} \times \text{Presences (0 - 100\%)}$$

Register:

1. Register the groups at:

https://docs.google.com/spreadsheets/d/10BxMIxyZZ7qSLq_LIJst0FSEJftXqN6KCkraBVngS4/edit#gid=0

2. Choose the participant/folder that your group will work on and indicate it on the form.

Milestone 1: 03/Mar/2024

1. Develop a code for importing audio signals. Note: you can use the **audioread** function of MATLAB.
2. Reproduce and plot an example of the imported signals, identifying the digit to which each one corresponds, and indicating the horizontal axis in seconds, as shown in the example of Figure 2.

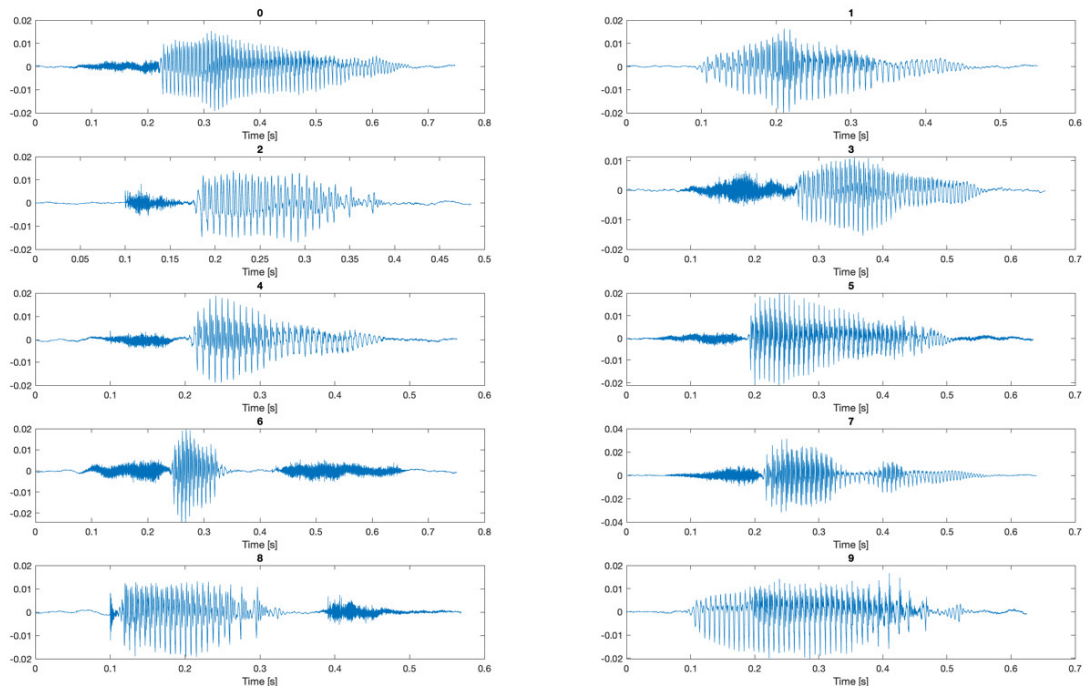


Fig. 2: Audio signals corresponding to digits from 0 to 9.

3. Visually identify and implement the calculation of possible temporal *features* of the signal that allow differentiating between digits, such as energy (total, per time intervals, differences between intervals, etc.), amplitude (maximum, minimum, amplitude ratio, standard deviation, etc.), or other temporal measures that you consider relevant.

Note: A strategy that can improve differentiation between digits is to perform a preprocessing step to ensure that all files have the same duration and amplitude range. For this, you can use the following strategies:

- Remove the initial "silence" from the signals to ensure that they all start exactly at the same time. This can be implemented through the analysis of Energy in time windows.
- Normalize the amplitude based on the maximum and minimum amplitude of the samples. This ensures that common sound recording problems (such as the distance from the person to the microphone) do not interfere with the analysis.
- Add (or remove) silence at the end of the files to ensure that they all have the same total duration.

4. Using the graphical representation of the temporal features extracted in the previous point, identify the three characteristics that allow a better discrimination between digits. For this, you can use *boxplots*, 2D, 3D graphs, etc. Some examples are presented in Figure 3.
Indicate the chosen characteristics.

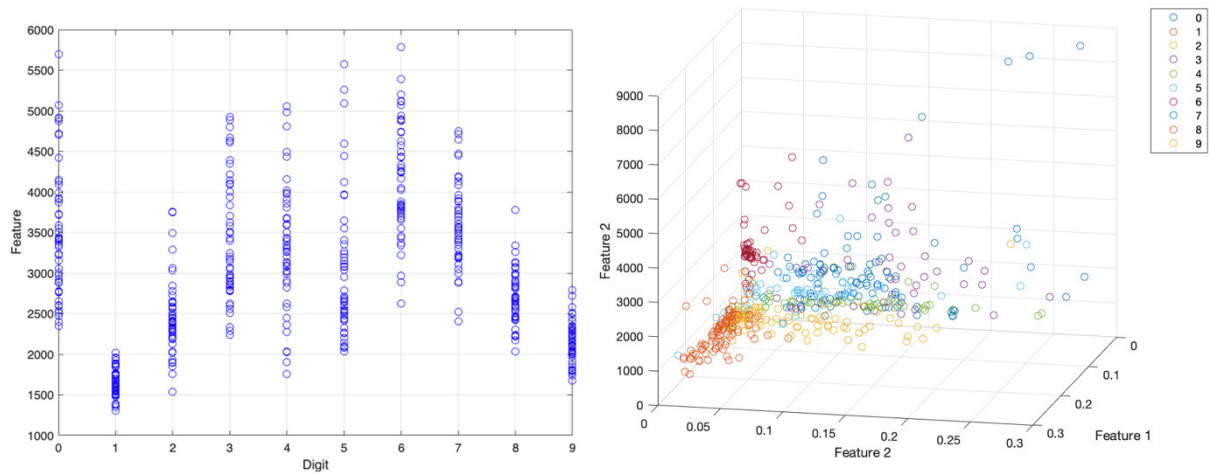


Fig. 3. Graphical representations of the extracted features.

Milestone 2: 8/Apr/2024

5. Compute, for each digit, the median amplitude spectrum, normalized by the number of samples (i.e., equivalent to the modulus of the coefficients of the complex Fourier series), and only for positive frequencies. Also, compute the first quartile (25%) and third quartile (75%). Figure 4 shows an example.

Note: you can use the **quantile** function in MATLAB.

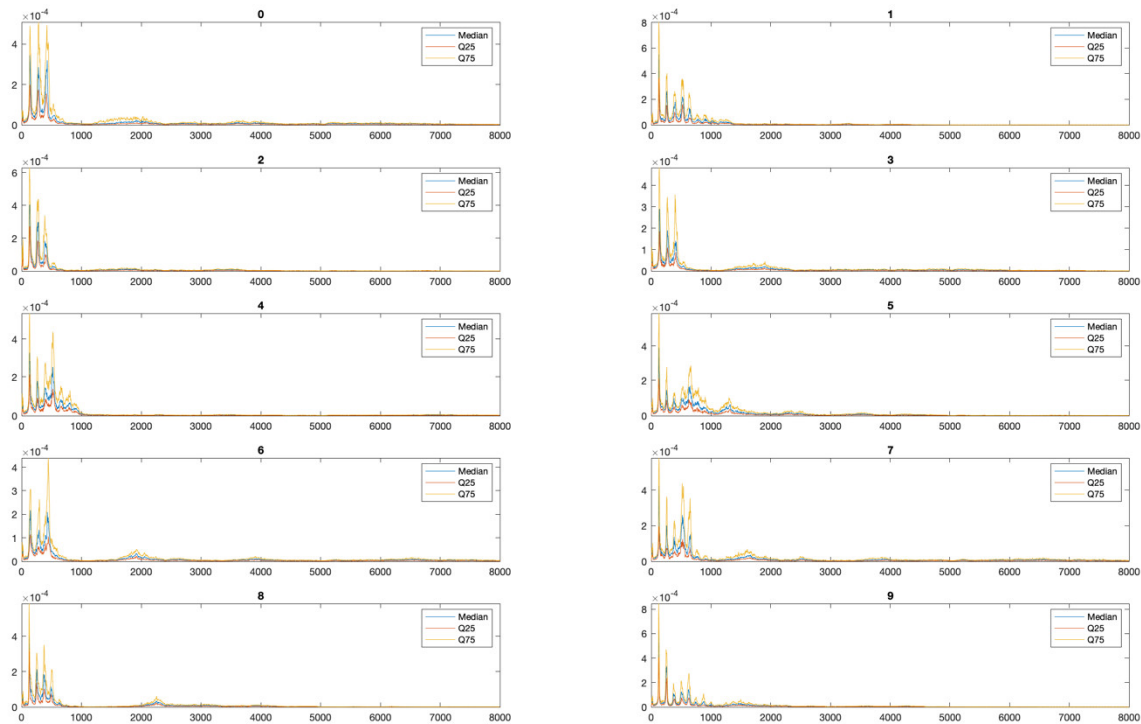


Fig. 4: Normalized median amplitude spectrum, first quartile, and third quartile for digits 0-9.

Note: for better visualization, the complete frequency axis is not shown.

6. Compare three different types of windows. What is the effect of the different windows? **Comment the results.**
7. Identify possible spectral features that allow differentiation between digits such as: spectral peaks (position and amplitude), spectral means, *spectral edge frequency*, etc.
8. Using the graphical representation of the spectral features extracted in the previous point, identify the three characteristics that allow better discrimination of the digits. For this, you can use *boxplots*, 2D, 3D graphs, etc. **Indicate the chosen characteristics.**

Milestone 3: 5/May/2024

9. Calculate, for each digit, the Short-Time Fourier Transform (STFT), as exemplified in Figure 5. Use different parameterizations, for example window size, overlap, number of points for FFT calculation, etc., and identify the parameters that seem best for the objective of the work.

Comment on the observations.

Note: use the `spectrogram` function in MATLAB (consult the associated documentation).

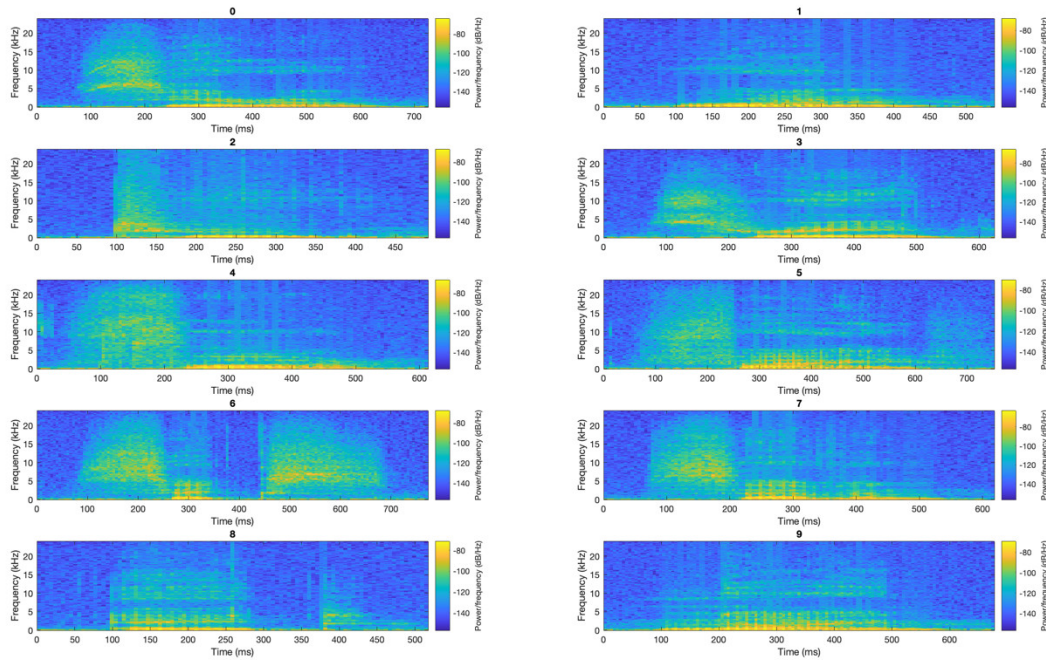


Fig. 5: STFT for the digits 0-9.

10. Extract power characteristics from different time-frequency windows that you find most relevant for differentiating the digits.
11. Using the graphical representation of the time-frequency characteristics extracted in the previous point, identify the three characteristics that allow better discrimination of the digits. For this, you can use *boxplots*, 2D, 3D graphs, etc. **Indicate the chosen characteristics.**

Milestone 4: 19/May/2024

12. Considering the best characteristics extracted from the time signals, frequency, and time-frequency domains in points 4, 8, and 11 respectively, define possible decision rules (of the *if ... else* type) that allow for better separation of the digits. For the definition of the rule set, you can use 5 to 9 different characteristics.
13. Compare the digits assigned to each audio based on the rules defined in the previous point with the actual digits and define the percentage of correct classifications as:

$$\text{Accuracy} = \frac{\text{Digits correctly identified}}{\text{Total number of digits}} \times 100$$

14. From the total set of characteristics, choose the three that showed the best results in digit differentiation and create a 3D plot of the data, as shown in Figure 3. **Comment on the observations.**